Lam Fung Academy · LF Academy

Primary 5 · Semester 2 · 23rd Lesson · Student Handout

Fraction mixed operations + four mixed operations

Unit three · Three fraction mix + four first after order · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 + Modern Education 5L B Unit 12

Core traps:□ T6 Wrong order of mixed operations · T9 Fractional operations - the numerator forgets to synchronize after the common division · The result is not reduced

SSPA association: 🔴 High frequency Paper 1 accounts for about 15-20%, Paper 2 accounts for about 10%

Prerequisite knowledge: Course 7-8 (Addition and subtraction of different denominators) · Course 9 (Multiplication of fractions) · Course 22 (Division of fractions)

Goal of this course:❶ Continuous addition and subtraction of three fractions ❷ Mixed multiplication and division of three fractions ❸ Four rules of order ❹ Comprehensive formula trap

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 4 question · 5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	Calculate LCM(3, 4, 6) = ? (least common multiple of three numbers)	Basic	
2	Calculation: $\frac{1}{2} + \frac{1}{3} =$? (Review addition of common fractions with different denominators)	Basic	
3	Calculation: $\frac{2}{3} \times \frac{1}{2} =$? (Review multiplication of fractions)	Basic	
4	Calculation: $\frac{3}{4} \div \frac{1}{2} =$? (Review fraction division: the fraction after ÷ needs to be reversed!)	Advanced	

II.Core Knowledge ＋ Worked Examples

Knowledge point 1: Adding and subtracting mixed three fractions 🔴 SSPA

① Add and subtract three or more fractions with different denominators →**First find the LCM of all denominators**

② Each fraction divides to the same denominator (the numerator is expanded simultaneously)

③ The denominator remains unchanged, and the numerator is calculated in the order of addition and subtraction

④ The result is reduced to the simplest fraction (improper fraction → mixed fraction)

⑤ **Note: LCM needs to find all denominators, not just two!**

Example 1

calculate: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$

Example 2

calculate: $\frac{3}{4} - \frac{1}{2} + \frac{1}{8} = ?$

Knowledge Point 1 Synchronization practice (must write out: ①Find the LCM that has/havedenominator ②find a common denominator ③numerator addsubtract in order ④simplify)

#	Question	Difficulty	Working Space

5	$\frac{1}{4} + \frac{1}{2} + \frac{1}{8} = ?$	🌱	
6	$\frac{2}{3} + \frac{1}{6} + \frac{1}{2} = ?$	🌱	
7	$\frac{5}{6} - \frac{1}{3} + \frac{1}{2} = ?$	🌿	
8	$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} = ?$	🌿	

⚠ **Warning: When adding and subtracting three fractions, the LCM must include all denominators. For example, the denominator is 2, 3, 4 → LCM(2,3,4)=12. If it is not LCM(2,3)=6, stop! 4 does not divide 6.**

Knowledge point 2: Mixed multiplication and division of third fractions 🔴 SSPA required test

Core rule: all division signs become multiplication signs (reverse divisors), then all numerators are multiplied ÷ all denominators are multiplied

① When encountering ÷ →, reverse the fraction after ÷, ÷ becomes ×

② All multiplications are calculated from left to right

③ You can **cross reduction**(The common factor of any numerator can be reduced by any denominator)

④ Finally, the numerator is multiplied by the denominator ÷ the denominator is multiplied → reduction

Original formula		÷ Change ×, reverse the fraction		
1/3 × 1/2 ÷ 1/6	→	**1/3 × 1/2 × 6/1**	=	**6/6 = 1**
÷ 1/6 (divide by a fraction) |||SEP|||	→	× 6/1（倒數） （乘以倒數）		口訣：「見除變乘，分數倒轉」

Trap detonation example 3 (the most important demonstration in this class)

calculate: $\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4} = ?$

❌ Common mistakes (70% students)

$$\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4} = \frac{2}{6} \div \frac{1}{4} = \frac{2}{6} \times \frac{1}{4}$$ ❌

Calculate the previous multiplication first, then forget to reverse when dividing, and mistakenly change ×½ into ×2 and then ×¼

✅ Correct solution

$$\frac{1}{2} \times \frac{2}{3} \times \frac{4}{1} = \frac{8}{6} = \frac{4}{3} = 1\frac{1}{3}$$

÷¼ → ×4/1, all ÷ become ×, and then reduce from left to right

🧠 **Tip: "Change all division signs into multiplication signs, and reverse the subsequent fractions; cross reduction is the most labor-saving, and the numerator and denominator should be equal."**

Example 4

calculate: $\frac{3}{4} \div \frac{1}{2} \times \frac{2}{3} = ?$

Knowledge Point 2 Synchronization practice (write out: ①÷→×reverse ②crosssimplify ③numerator×÷denominator× ④simplify)

#	Question	Difficulty	Working Space
9	$\frac{1}{3} \times \frac{1}{2} \div \frac{1}{6} = ?$	🌱	
10	$\frac{2}{5} \div \frac{2}{3} \times \frac{1}{2} = ?$	🌿	
11	$\frac{3}{4} \times \frac{2}{3} \div \frac{1}{2} = ?$	🌿	
12	$\frac{5}{6} \div \frac{2}{3} \times \frac{1}{2} = ?$	🌿	

⚠️ High-frequency errors: When mixing multiplication and division, if you do multiplication first and then division, you may make an error. The best strategy: first turn all ÷ into × (reverse the divisors) and process them uniformly.

Knowledge Point 3: Four Mixed Operation Sequences 🔴 SSPA Required Exam

Operation precedence (from high to low):

① **Parentheses ( )**— the calculations inside the parentheses are evaluated first

② **Formula: parentheses first, then multiplication and division, and finally addition and subtraction**(from left to right at the same level)

③ **+ and −**— Addition and subtraction last (from left to right)

④ **Formula: brackets first, then multiplication and division, and finally addition and subtraction**(Same level from left to right)

①	②	③	④

Parentheses first	Multiplication and division take precedence	Addition and subtraction last	Same level left to right
The formula within ( ) must be calculated first	× ÷ is calculated before + −	+ − after parentheses and multiplication and division	Operations at the same level are performed from left to right

Trap detonation example 5

calculate: $\frac{1}{2} + \frac{1}{3} \times \frac{1}{4} = ?$

❌ Common Mistake — Doing it Directly from Left to Right

$(\frac{1}{2} + \frac{1}{3}) \times \frac{1}{4} = \frac{5}{6} \times \frac{1}{4} = \frac{5}{24}$

Treat + as if it were on the same level as

✅ Correct solution: first × then +

$\frac{1}{2} + \frac{1}{12} = \frac{7}{12}$

Calculate first ⅓×¼=1/12, then add ½=6/12 → 7/12

Example 6

Calculation: $(\frac{1}{2} + \frac{1}{3}) \div \frac{1}{6} = ?$ (Note: Calculate first in parentheses, then ÷)

Knowledge Point 3 Synchronization practice (must mark the calculate order: ①brackets → ②×÷ → ③+−)

#	Question	Difficulty	Working Space
13	$\frac{1}{2} + \frac{1}{3} \div \frac{1}{6} = ?$	🌿	
14	$\frac{2}{3} \times (\frac{1}{2} + \frac{1}{4}) = ?$	🌿	
15	$\frac{3}{4} - \frac{1}{2} \times \frac{1}{3} = ?$	🌿	

🧠 Tip: "Put parentheses first, then multiplication and division, and finally addition and subtraction; the same level is from left to right, and the order cannot be messed up."

Knowledge Point 4: Comprehensive Application - Four Mixed Text Questions 🔴 SSPA Required

Key to solving the problem:① **Understand the question**→ Determine which operations are used ②**Column comprehensive calculation**(note the brackets and order)

③ **Calculate step by step in the four order** ④ **Write a complete answer**(Step points will be deducted if you do not write an answer!)

Example 7

Xiao Ming has $\frac{1}{2}$ cakes, and his mother buys twice as many $\frac{1}{3}$ cakes. What fraction of the cake does Xiao Ming have now?

Example 8

A rope is $\frac{3}{4}$ meters long. After using $\frac{1}{2}$ meters, the remaining $\frac{2}{3}$ of the rope is used for handicrafts. How much rice did you use to make it?

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
16	Xiaomei has $\frac{2}{5}$ liters of orange juice. After drinking $\frac{1}{10}$ liters, she divides the remainder into 3 glasses. How many liters are there in each cup?	🌳	
17	One piece of $\frac{1}{3}$ was used as a gift, and the remaining $\frac{1}{2}$ was used for decoration. What fraction of the total rope does the decorative part account for?	🌳	
18	The younger brother's pocket money belongs to the older brother |||SEP|||, and the younger brother's pocket money belongs to the younger brother|||SEP|||. What fraction of my sister’s pocket money is that of my brother’s? $\frac{1}{2}$, the sister’s is the brother’s $\frac{2}{3}$. What fraction of my sister’s pocket money is that of my brother’s?	🌳	

III. Lesson Layered Synchronization Practice

Basic layer (total 4 questions, everyone must do)

#	Question	Difficulty	Working Space
19	$\frac{1}{3}+\frac{1}{6}+\frac{1}{2}=?$	🌱	

20	$\frac{5}{8} - \frac{1}{4} + \frac{1}{2} = ?$	🌱	
21	$\frac{1}{4} \times \frac{2}{3} \div \frac{1}{2} = ?$	🌱	
22	$\frac{1}{2} + \frac{1}{4} \div \frac{1}{2} = ?$	🌿	

Advanced layer (total 3 questions, 🧍🚀 choose do)

#	Question	Difficulty	Working Space
23	$\frac{3}{5} + \frac{1}{2} - \frac{1}{10} + \frac{1}{5}$=? (Four fractions addition and subtraction mixed)	🌿	
24	$\frac{7}{8} \div \frac{3}{4} \times \frac{1}{2} = ?$	🌿	
25	$(\frac{1}{2} - \frac{1}{3}) \div \frac{1}{6} + \frac{1}{4} = ?$	🌳	

🌳 challenge layer (total 2 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
26	$\frac{1}{2} + \frac{2}{3} \times \frac{3}{4} \div \frac{1}{2}$=? (Four mixing rules: first ×÷, then +−)	🌳	
27	$\frac{2}{3} \div \frac{1}{2} \times (\frac{1}{2} + \frac{1}{4}) = ?$ (Parentheses first, then multiplication and division from left to right)	🌳	

IV.🏔 Ultimate challenge area

#	Question	Difficulty	Working Space
🏔 1	Calculation: (|||SEP||| = ? (two brackets, then ×÷ - pay attention to the order!)$\frac{1}{2} + \frac{1}{3}) \times (\frac{2}{3} - \frac{1}{4}) \div \frac{1}{6}$=? (Two parentheses, then ×÷ - pay attention to the order!)	🏔	
🏔 2	Bottle of Juice$\frac{3}{4}$Liters. I drank liters of$\frac{1}{6}$at breakfast and the remaining $\frac{2}{3}$at lunch. How many liters are left after drinking lunch? (Calculation using comprehensive formulas)	🏔	

Advanced mixed question (🧗🚀 choose do)

#	Question	Difficulty	Working Space
M1	$\frac{1}{2}+\frac{1}{3}\div\frac{1}{6}-\frac{1}{4}$=? (÷ → − → +, pay attention to the order!)	🌳	
M2	$\frac{2}{3}+\frac{1}{2}\times(\frac{1}{3}+\frac{1}{6})$ = ? (Brackets first, then ×, and finally +)	🌳	

V. Class afterhomework

Basic must-doquestions (total 6 questions, mustwrite outcalculate order mark)

#	Question	Difficulty	Working Space
H1	$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}=?$	🌱	
H2	$\frac{3}{5}+\frac{1}{10}+\frac{1}{2}=?$	🌱	
H3	$\frac{2}{3}\times\frac{1}{4}\div\frac{1}{2}=?$	🌱	
H4	$\frac{3}{4}-\frac{1}{8}+\frac{1}{2}=?$	🌿	
H5	$\frac{1}{2}+\frac{1}{3}\times\frac{1}{4}=?$	🌿	
H6	$\frac{3}{5}+\frac{1}{2}\times\frac{2}{3}=?$	🌿	

For advanced, choose doquestion (total 2 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H7	$\frac{2}{3}\div\frac{1}{2}+\frac{1}{4}\times\frac{2}{3}=?$	🌳	
H8	A ribbon is $1\frac{1}{2}$meter long. After using |||SEP|||, the remaining $\frac{1}{3}$is used to wrap gifts. How much rice did it take to wrap the gift? $\frac{1}{2}$Used to wrap gifts. How much rice did it take to wrap the gift?	⛰️	

VI. The Lessoncorecommon errorsummary

✅ Self-examination in this hall (tick after completion)

- ☐ I know the pitfalls of solving each knowledge point
- ☐ I can complete 🌱basic questions independently
- ☐ I can challenge 🌿advanced questions
- ☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

- ☐ Identify all trap types in our hall
- ☐ Solve 🌱basic questions independently (100% correct)
- ☐ Challenge 🌿Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

✅ 本堂自我檢查（完成後打剔）

- ☐ 我識得分辨每個知識點嘅陷阱
- ☐ 我能夠獨立完成🌱基礎題
- ☐ 我能夠挑戰🌿進階題
- ☐ 我記得住口訣

#	common error	Correct Approach
1	**Three fractions LCM only finds two |||SEP|||: denominator 2,3,4 only finds LCM(2,3)=6**: Denominator 2,3,4 only looks for LCM(2,3)=6	LCM must include all denominators! LCM(2,3,4)=12
2	**When mixing multiplication and division, forget to change ÷ into ×** : $\frac{1}{2}\times\frac{1}{3}\div\frac{1}{6}$ and only count the first two	All ÷ become × (the divisor is reversed) and are processed uniformly
3	**Four order errors—adding and subtracting first, then multiplying and dividing**	First ×÷ and then +−, if there are parentheses, it is calculated first
4	**Uncommon points in brackets** : $(\frac{1}{2}+\frac{1}{3})$ Write directly $\frac{2}{5}$	The scores in parentheses must be divided first and then calculated!
5	**During cross reduction, only the same denominator group is reduced**	The common factors of any numerator can be reduced by any denominator
6	**The result is not reduced/improper fraction is not transferred**	The final step must be to check the reduction and transfer of mixed fractions
7	**The order of expressions in application questions is reflected incorrectly**(missing brackets)	Check when formulating: What does the question ask you to do first? Do I need to use parentheses?

🏠 家長角落 · Parent Corner

今日學咗咩？ 小朋友學咗「**Fraction mixed operations + four mixed operations**」。

最易錯嘅 3 個陷阱： 留意老師圈出嘅陷阱題目

你可以問小朋友：「你可唔可以解釋「Fraction mixed operations + four mixed operations」嘅最重要口訣俾我聽？」

溫馨提示：唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

📝 **教師參考**：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號：🌳挑戰層 17-21